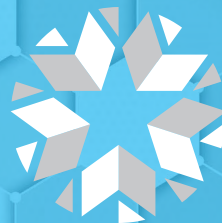


OKLAHOMA SCHOOL TESTING PROGRAM

TEST BLUEPRINT AND ITEM SPECIFICATIONS
GRADE 3 MATHEMATICS



OKLAHOMA
Education

1 The distance between Washington, D.C., and Oklahoma City is about one thousand, three hundred, twenty miles. How is this distance written in numerals?

- A** 132 miles
- B** 1,032 miles
- C** 1,302 miles
- D** 1,320 miles

Standard: 3.N.1.1 Read, write, discuss, and represent whole numbers up to 100,000. Representations should include but are not limited to numerals, words, pictures, number lines, and manipulatives (e.g., $350 = 3 \text{ hundreds}, 5 \text{ tens} = 35 \text{ tens} = 3 \text{ hundreds}, 4 \text{ tens}, 10 \text{ ones}$).

Depth-of-Knowledge: 1

This item is a DOK 1 because it requires the student to perform a simple procedure, representing a number given in words as a number using numerals.

Distractor Rationale

- A. The student did not account for the 0 in 20.
- B. The student confused 320 with 32.
- C. The student confused 320 with 302.
- D. Correct. The student demonstrated an ability to write a number up to 100,000.**

2 A polar bear weighed $2,000 + 200 + 9$ pounds.
Which number is the same as the weight of this polar bear?

- A** 229 pounds
- B** 2,209 pounds
- C** 2,290 pounds
- D** 22,009 pounds

Standard: 3.N.1.2 Use place value to describe whole numbers between 1,000 and 100,000 in terms of ten thousands, thousands, hundreds, tens and ones, including written, standard, and expanded forms.

Depth-of-Knowledge: 1

This item is a DOK 1 because it requires the student to perform the simple procedure of changing a number given in expanded form into standard form.

Distractor Rationale

- A. The student made a place value error.
- B. Correct. The student demonstrated an ability to translate a number given in expanded form into standard form.**
- C. The student confused 9 and 90.
- D. The student made a place value error.

- 3** An airplane flew a total of 38,215 miles last week. This week the plane will fly 10,000 miles fewer than it did last week.

What is the distance the airplane will fly this week, in miles?

- A** 28,215 miles
- B** 37,215 miles
- C** 39,215 miles
- D** 48,215 miles

Standard: 3.N.1.3 Applying knowledge of place values, use mental strategies (no written computations) to find 100 more or 100 less than a given number, 1,000 more or 1,000 less than a given number, and 10,000 more or 10,000 less than a given number, up to a five-digit number.

Depth-of-Knowledge: 1

This item is a DOK 1 because it requires the student to perform a simple procedure, finding 10,000 less than a given number.

Distractor Rationale

- A. Correct. The student demonstrated an ability to find 10,000 less than a given five-digit number.
- B. The student confused 10,000 and 1,000.
- C. The student confused 10,000 and 1,000 and found 1,000 more instead of 10,000 fewer.
- D. The student found 10,000 more instead of 10,000 fewer.

- 4** Which number could be placed in the blank to make the number sentence true?

$$5,426 > \underline{\quad}$$

- A** 5,430
- B** 5,617
- C** 5,584
- D** 5,418

Standard: 3.N.1.4 Use place value to compare and order whole numbers up to 100,000, using comparative language, numbers, and symbols.

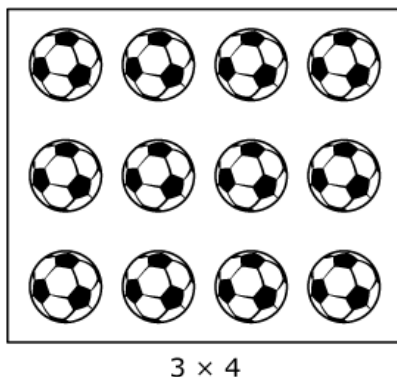
Depth-of-Knowledge: 1

This item is a DOK 1 because it requires the student to perform a simple procedure, comparing two numbers.

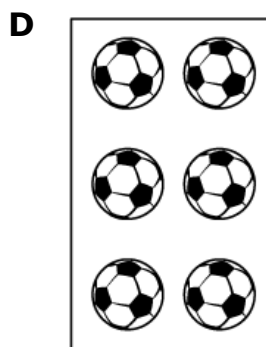
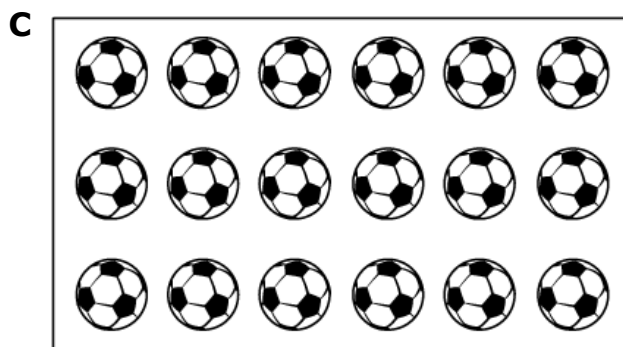
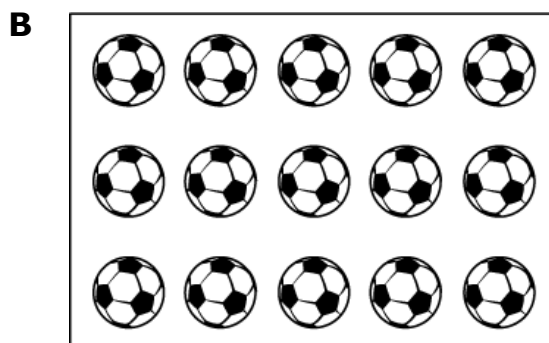
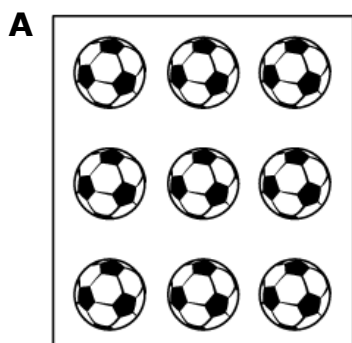
Distractor Rationale

- A. The student confused the greater than and less than symbols or compared the ones places only.
- B. The student confused the greater than and less than symbols or compared the tens places only.
- C. The student confused the greater than and less than symbols or compared the ones places only.
- D. Correct. The student demonstrated an ability to use place value to compare numbers up to 100,000.

5 This array shows 3×4 .



Which array shows 3×6 ?



Standard: 3.N.2.1 Represent multiplication facts by modeling a variety of approaches (e.g., manipulatives, repeated addition, equal-sized groups, arrays, area models, equal jumps on a number line, skip counting).

Depth-of-Knowledge: 2

This item is a DOK 2 because it requires the student to use the given model to identify a model for a different multiplication expression.

Distractor Rationale:

A. The student confused 3×6 and 3×3 .

B. The student confused 3×6 and 3×5 .

C. Correct. The student demonstrated an ability to represent a multiplication fact by using an array.

D. The student saw 3 groups and a total of 6 soccer balls.

- 6** During gym class, 6 students took turns jumping rope. Each student jumped rope for 7 minutes.

What is the total number of minutes these 6 students spent jumping rope?

- A** 13 minutes
- B** 36 minutes
- C** 42 minutes
- D** 49 minutes

Standard: 3.N.2.2 Demonstrate fluency with multiplication facts using factors up to 10.

Depth-of-Knowledge: 2

This item is a DOK 2 because it requires the student to determine that this is a multiplication scenario and then to compute the total.

Distractor Rationale:

- A. The student added the two given numbers.
- B. The student made a fact error.
- C. Correct. The student demonstrated an ability to fluently multiply with factors up to 10.**
- D. The student made a fact error.

- 7** This table shows the number of books checked out from the school library on three days last week.

Books Checked Out

Day	Number of Books Checked Out
Monday	241
Tuesday	121
Wednesday	134

How many more books were checked out on Monday than on Wednesday?

- A** 120
- B** 117
- C** 113
- D** 107

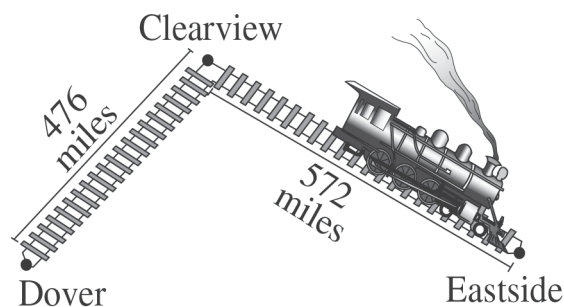
Standard: 3.N.2.3 Use strategies and algorithms based on knowledge of place value and equality to fluently add and subtract up to five-digit numbers (answer not to exceed 100,000).

Depth-of-Knowledge: 2

This item is a DOK 2 because it requires the student to extract the necessary information from the table, determine that this is a subtraction scenario and then to compute the difference.

Distractor Rationale:

- A. The student compared Monday to Tuesday.
- B. The student failed to reduce the tens by one when regrouping from the tens to the ones.
- C. The student subtracted the smaller digit from the larger digit in both places.
- D. Correct. The student demonstrated an ability to subtract three-digit numbers.**



To the nearest ten, about how far does the train travel from Dover to Eastside?

- A** 950 miles
- B** 1050 miles
- C** 1060 miles
- D** 1150 miles

Standard: 3.N.2.4 Recognize when to round numbers and apply understanding to estimate sums and differences to the nearest ten thousand, thousand, hundred, and ten.

Depth-of-Knowledge: 2

This item is a DOK 2 because it requires the student to first make a decision about how to approach the problem. The student needs to both add and round to the nearest ten.

Distractor Rationale:

- A. The student added only the numbers in the hundreds place.
- B. Correct. The student demonstrated an ability to round numbers to the nearest ten to estimate a sum.**
- C. The student rounded 572 incorrectly.
- D. The student estimated a distance too high.

9 Three elephants at a zoo weigh a total of 9,898 pounds. One elephant weighs 7,859 pounds. Another elephant weighs 1,602 pounds. How many pounds does the third elephant weigh?

- A** 437 pounds
- B** 1,447 pounds
- C** 1,641 pounds
- D** 2,263 pounds

Standard: 3.N.2.5 Use addition and subtraction to solve problems involving whole numbers. Use various strategies, including the relationship between addition and subtraction and the context of the problem to assess the reasonableness of results.

Depth-of-Knowledge: 2

This item is a DOK 2 because it requires the student to first make a decision about how to approach the problem. The student needs to perform more than one calculation.

Distractor Rationale:

- A. Correct.** The student demonstrated an ability to use addition and subtraction to solve a real-world problem involving whole numbers.
- B.** The student made a computation error.
- C.** The student made a computation error.
- D.** The student selected the answer closest to $9,898 - 7,859$.

10 What is the value of the fraction $\frac{5}{8}$?

- A** one fifth
- B** eight fifths
- C** one eighth
- D** five eighths

Standard: 3.N.3.1 Read and write fractions with words and symbols using appropriate terminology (i.e., numerator and denominator).

Depth-of-Knowledge: 1

This item is a DOK 1 because it requires the student to perform a simple procedure, representing a fraction given in numerals as a fraction using words.

Distractor Rationale:

- A. The student focused on the number in the numerator.
- B. The student reversed the numerator and the denominator.
- C. The student focused on the number in the denominator.
- D. **Correct.** The student demonstrated an ability to read fractions with words and symbols.

11

This number line is divided into equal parts. Place the dot on the number line to represent the fraction $\frac{5}{8}$. To place the dot, click and drag it to the correct location on the number line. To change the location of the dot, click and hold it, and then drag it to the new location.



Standard: 3.N.3.2 Model fractions using length, set, and area for halves, thirds, fourths, sixths, and eighths.

Depth-of-Knowledge: 1

This item is a DOK 1 because it requires the student to recall the simple procedure of how to represent a fraction on a number line.

Correct



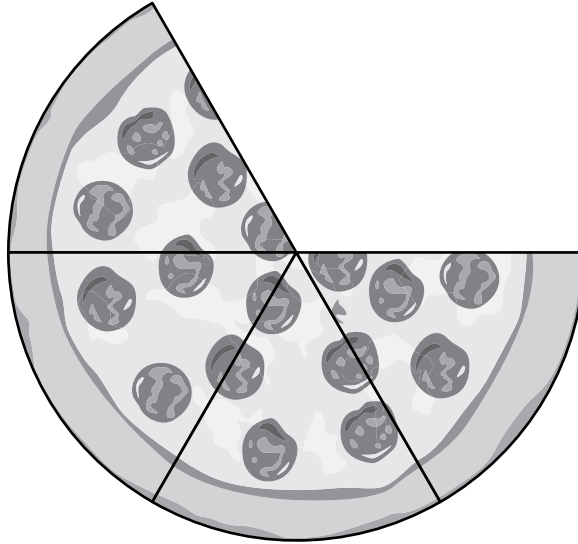
Incorrect



Distractor Rationale:

The student started counting the tic marks at 0 and then found the fifth tic mark.

- 12** Joe cut a pizza into 6 equal pieces. He ate 2 pieces.



What fraction of the pizza is left?

- A** $\frac{2}{4}$
- B** $\frac{4}{6}$
- C** $\frac{6}{4}$
- D** $\frac{4}{2}$

Standard: 3.N.3.3 Apply understanding of unit fractions and use this understanding to compose and decompose fractions related to the same whole.

Depth-of-Knowledge: 2

This item is a DOK 2 because it requires the student to use a visual model to translate into a numerical answer.

Distractor Rationale:

- A. The student used the number of pieces Joe ate as the numerator and the number of pieces left as the denominator.
- B. Correct. The student demonstrated an ability to use the numerator to describe the number of parts and the denominator to describe the number of partitions.**
- C. The student confused the numerator and the denominator.
- D. The student used the number of pieces left as the numerator and the number of pieces Joe ate as the denominator.

13 Jorge has collected these bills.



Which number sentence shows the total value of the bills?

- A** $\$10 + \$3 = \$13$
- B** $\$13 + \$3 = \$16$
- C** $\$15 + \$3 = \$18$
- D** $\$18 + \$3 = \$21$

Standard: 3.N.4.2 Add and subtract a collection of bills up to twenty dollars using whole dollars in monetary transactions.

Depth-of-Knowledge: 2

This item is a DOK 2 because it requires the student to represent the amount of money shown using a number sentence in which some of the bills are combined.

Distractor Rationale:

- A. The student forgot the \$5 bill.
- B. The student forgot the \$5 bill and added the three \$1 bills twice.
- C. **Correct.** The student demonstrated an ability to use addition to determine the value of a collection of bills up to twenty dollars.
- D. The student added the three \$1 bills twice.

- 14** Connie is learning to play 15 songs on the piano. The table shows the number of songs Connie has left to learn at the end of each month.

Songs for Connie to Learn

Month	Number of Songs
January	15
February	13
March	11
April	9
May	?

Connie learns the same number of songs each month. How many songs will Connie have left to learn at the end of May?

- A** 2 songs
- B** 6 songs
- C** 7 songs
- D** 8 songs

Standard: 3.A.1.1 Create, describe, and extend patterns involving addition, subtraction, or multiplication to solve problems in a variety of contexts.

Depth-of-Knowledge: 2

This item is a DOK 2 because it requires the student to figure out the pattern shown in the table and then identify the number of songs for May.

Distractor Rationale:

- A. The student found that the numbers in the table decrease by 2 each month and did not know how to use this information.
- B. The student subtracted the largest number in the table from the smallest number.
- C. Correct. The student demonstrated an understanding of how to extend a pattern.**
- D. The student thought that May should be 1 less than April.

Input	Output
12	5
19	12
25	18

Which rule could be used to change the input number to the output number in the table above?

- A** divide by 7
- B** multiply by 6
- C** add 6
- D** subtract 7

Standard: 3.A.1.2 Describe the rule (limited to a single operation) for a pattern from an input/output table or function machine involving addition, subtraction, or multiplication.

Depth-of-Knowledge: 2

This item is a DOK 2 because it requires the student to determine the rule shown in an input/output table and then describe it using mathematical terms.

Distractor Rationale:

- A. The student chose the wrong operation.
- B. Balance distractor
- C. The student saw 12 and 18 in the output and applied the rule $12 + 6 = 18$.
- D. **Correct.** The student demonstrated an ability to describe the rule for a pattern from an input/output table.

- 16** Seth wants to visit all 50 states. He has visited 14 states. The number sentence shows s , the number of states Seth has left to visit.

$$s + 14 = 50$$

How many states does Seth have left to visit?

- A** 36
- B** 44
- C** 46
- D** 64

Standard: 3.A.2.1 Use number sense with the properties of addition, subtraction, and multiplication, to find unknowns (represented by symbols) in one-step equations. Generate real-world situations to represent number sentences.

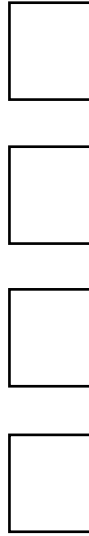
Depth-of-Knowledge: 2

This item is a DOK 2 because it requires the student to employ a strategy for finding the missing addend.

Distractor Rationale:

- A. Correct. The student demonstrated an ability to find an unknown represented by a symbol in an arithmetic problem by solving a one-step equation.
- B. Balance distractor
- C. The student computed $50 - 4$.
- D. The student added instead of subtracting.

17 Joey put pictures on his locker door as shown.



Which number sentence shows how many pictures Joey put on his locker door?

- A** $4 \times 1 = 4$
- B** $4 \times 0 = 4$
- C** $4 + 1 = 4$
- D** $4 \div 0 = 4$

Standard: 3.A.2.2 Identify, represent, and apply the number properties (commutative, identity, and associative properties of addition and multiplication) using models and manipulatives to solve problems.

Depth-of-Knowledge: 1

This item is a DOK 1 because it requires the student to perform a simple procedure, identifying the number sentence that represents a simple model.

Distractor Rationale:

- A. Correct.** The student demonstrated an ability to represent the identity property of multiplication using a model.
- B.** The student confused 0 and 1.
- C.** The student confused multiplication and addition.
- D.** The student chose an incorrect operation.

18 Which expression equals $(4 \times 2) \times 3$?

- A** $(4 + 2) + 3$
- B** $(2 \times 4) + 3$
- C** $4 \times (2 \times 3)$
- D** $2 + (4 \times 3)$

Standard: 3.A.2.2 Identify, represent, and apply the number properties (commutative, identity, and associative properties of addition and multiplication) using models and manipulatives to solve problems.

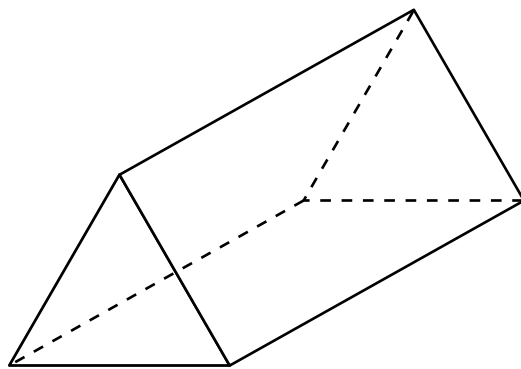
Depth-of-Knowledge: 1

This item is a DOK 1 because it requires the student to recall definitions, identifying equal expressions using the associative property of multiplication.

Distractor Rationale:

- A. The student thought multiplication and addition can be interchanged or chose because the order of the numbers is the same as in the original expression.
- B. The student knew the order of the factors can be changed when multiplying, but thought multiplication and addition can be interchanged.
- C. Correct. The student demonstrated an ability to apply the associative property of multiplication.**
- D. The student knew the order of the factors can be changed when multiplying, but thought multiplication and addition can be interchanged.

19



What is the total number of triangular faces on the figure?

- A** 2
- B** 3
- C** 5
- D** 6

Standard: 3.GM.1.1 Sort three-dimensional shapes based on attributes.

Depth-of-Knowledge: 1

This item is a DOK 1 because it requires the student to identify a fact, identifying the total number of triangular faces on a given figure.

Distractor Rationale:

- A. Correct.** The student demonstrated an understanding of the number of triangular faces on a triangular prism.
- B.** The student chose because triangles have 3 sides.
- C.** The student gave the total number of faces.
- D.** The student chose the number of edges at the bases.

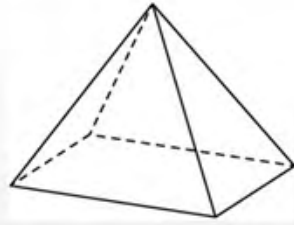
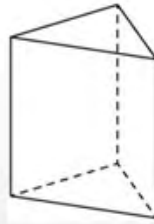
Match each sentence on the left to the figure it describes on the right. Each figure will have two sentences matched to it. Click one sentence on the left and then click the figure it describes on the right. To remove a line, hold the pointer over the line until it turns red, and then click it.

The figure has 2 bases.

The figure has exactly 5 vertices.

The figure has a triangular base.

The figure has a rectangular base.



Standard: 3.GM.1.1 Sort three-dimensional shapes based on attributes.

Depth-of-Knowledge: 2

This item is a DOK 2 because it requires the student to compare and contrast the features of two different shapes.

Distractor Rationale:

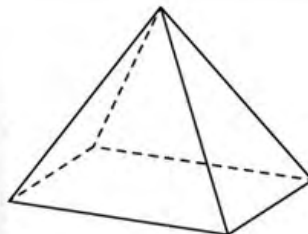
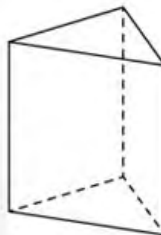
Correct

The figure has 2 bases.

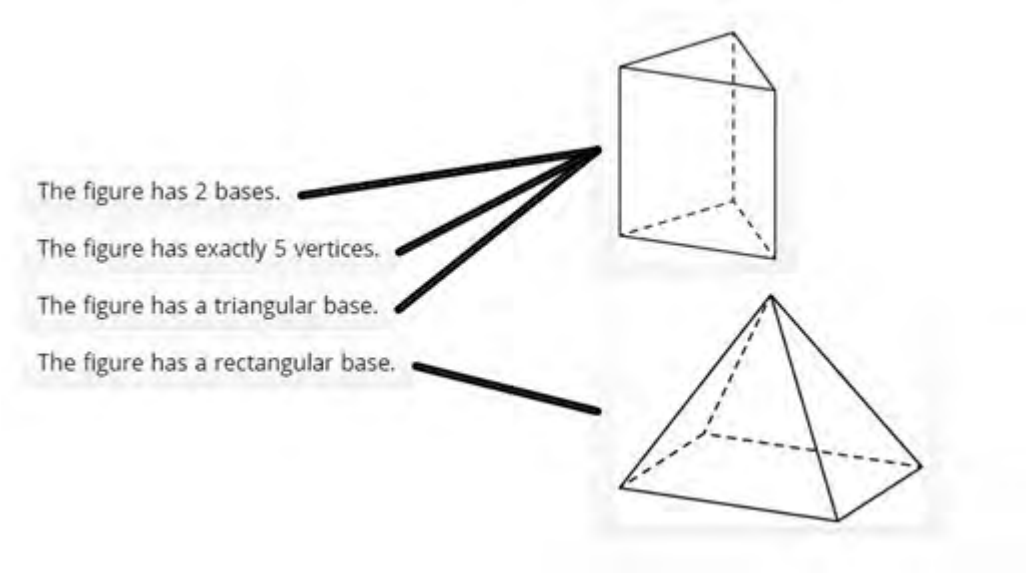
The figure has exactly 5 vertices.

The figure has a triangular base.

The figure has a rectangular base.

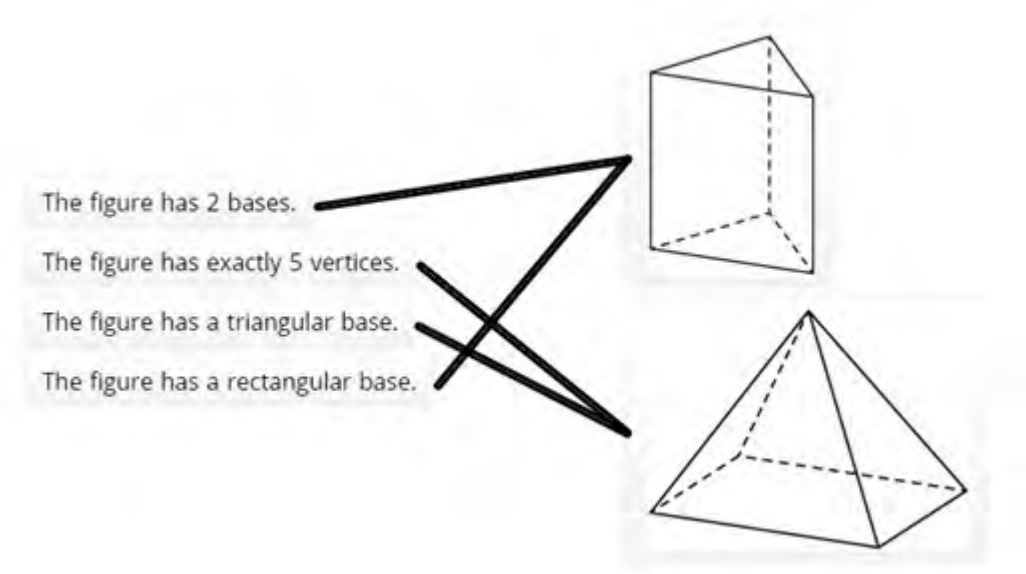


Incorrect



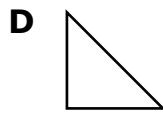
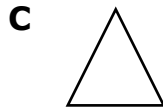
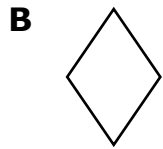
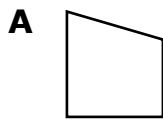
The student confused vertices and faces.

Incorrect



The student chose a rectangular base for the top shape because that is the forward-facing face and chose a triangular base for the bottom shape because that is the forward-facing face.

21 Which shape below appears to contain at least one acute, one obtuse, and one right angle?



Standard: 3.GM.1.3 Classify angles within a polygon as acute, right, obtuse, and straight.

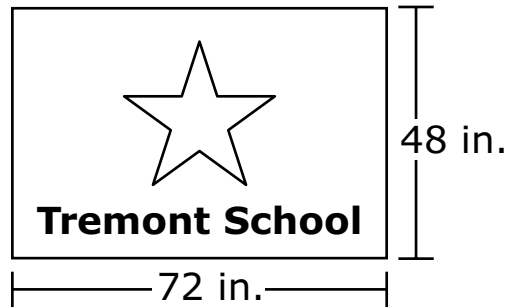
Depth-of-Knowledge: 2

This item is a DOK 2 because it requires the student to look at each shape and then identify the shape with the three identified angles.

Distractor Rationale:

- A. **Correct.** The student demonstrated an ability to classify angles as acute, right, and obtuse.
- B. The student thought one of the obtuse angles was a right angle.
- C. Balance distractor
- D. The student thought one of the acute angles was an obtuse angle.

- 22** This rectangular flag is flown outside Tremont School.



What is the perimeter of the flag?

- A** 96 inches
- B** 120 inches
- C** 144 inches
- D** 240 inches

Standard: 3.GM.2.1 Find the perimeter of a polygon, given whole number lengths of the sides, using a variety of models.

Depth-of-Knowledge: 2

This item is a DOK 2 because it requires the student to know what the perimeter means, and then use a strategy to find the perimeter of a rectangle with only two sides labeled.

Distractor Rationale:

- A. The student computed $48 + 48$.
- B. The student computed $48 + 72$.
- C. The student computed $72 + 72$.
- D. Correct. The student demonstrated an ability to find the perimeter of a rectangle.

23 Mrs. Steinberg’s class made a design using square pieces of paper. Each piece of paper was 1 foot wide by 1 foot long. The design was a rectangle, 5 feet wide by 7 feet long. How many square pieces of paper were used to make the design?

- A** 12 pieces of paper
- B** 20 pieces of paper
- C** 24 pieces of paper
- D** 35 pieces of paper

Standard: 3.GM.2.2 Analyze why length and width are multiplied to find the area of a rectangle by decomposing the rectangle into one unit by one unit squares and viewing these as rows and columns to determine the area.

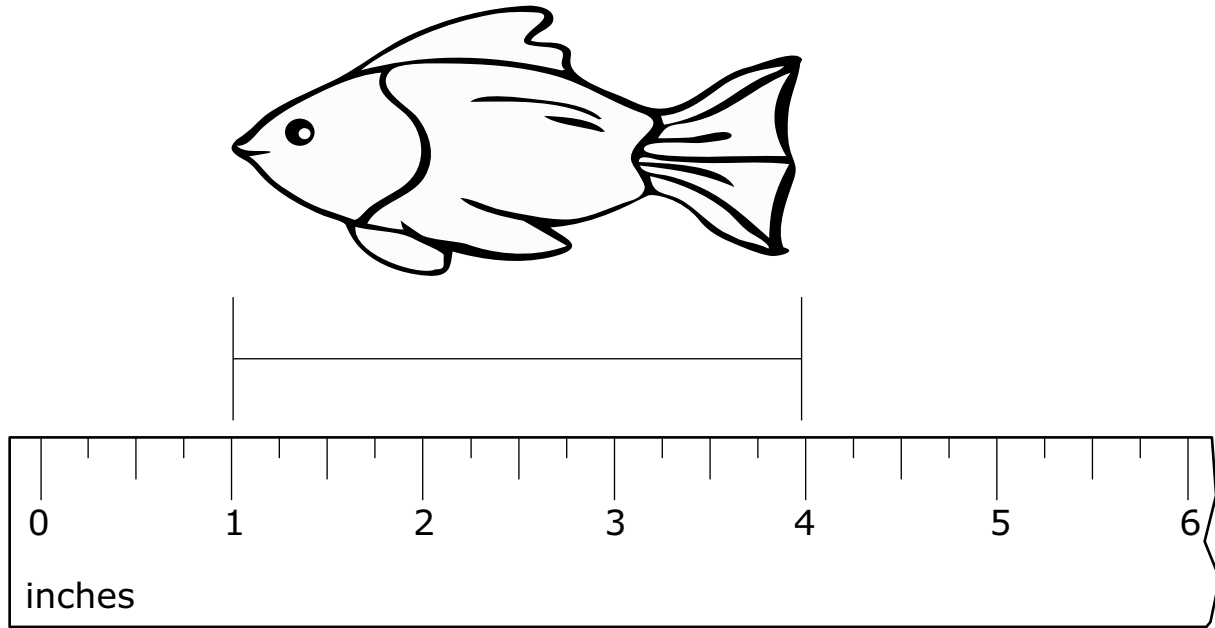
Depth-of-Knowledge: 3

This item is a DOK 3 because the student must use reasoning to determine the correct answer, identifying the area and then translating this into the number of pieces of paper.

Distractor Rationale:

- A. The student added the 5 and 7.
- B. Balance distractor
- C. The student computed $5+7+5+7$, finding the perimeter.
- D. Correct. The student demonstrated an ability to analyze why length and width are multiplied to find the area of a rectangle.

24 Jessie measured her goldfish as shown.



What is the length of Jessie's goldfish?

- A** 1 inch
- B** 3 inches
- C** 4 inches
- D** 6 inches

Standard: 3.GM.2.6 Choose an appropriate measurement instrument and measure the length of objects to the nearest whole yard, whole foot, or half inch.

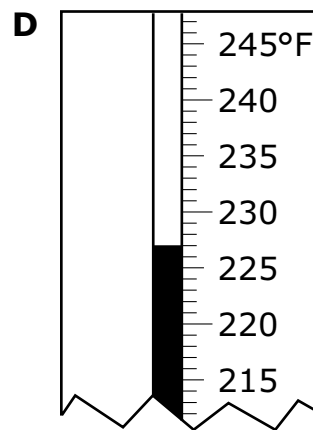
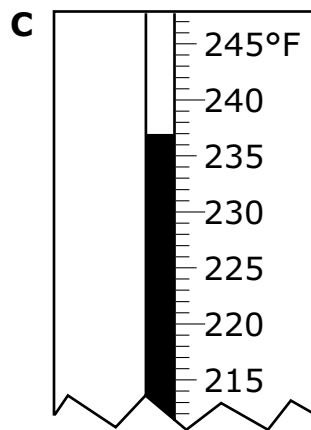
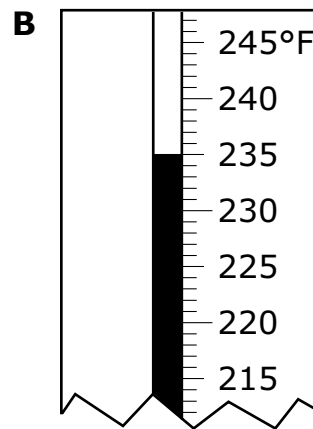
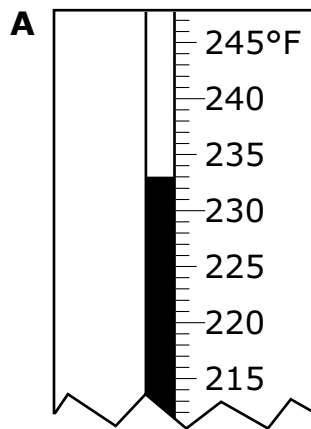
Depth-of-Knowledge: 1

This item is a DOK 1 because it requires the student to perform a simple procedure, measuring using a ruler.

Distractor Rationale:

- A. The student chose the number of inches for the starting point of the fish.
- B. Correct. The student demonstrated an ability to measure the length of an object to the nearest half inch.**
- C. The student chose the number of inches for the ending point of the fish.
- D. The student chose the last number shown on the ruler.

- 25** Hiram is making candy. He needs to cook the candy mixture to 237°F . Which thermometer reading is closest to this temperature?



Standard: 3.GM.2.7 Use an analog thermometer to determine temperature to the nearest degree in Fahrenheit and Celsius.

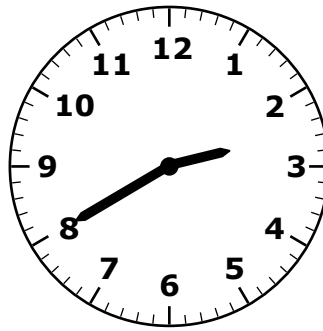
Depth-of-Knowledge: 1

This item is a DOK 1 because it requires the student to perform a simple procedure, identifying a temperature shown on a thermometer.

Distractor Rationale:

- A. The student moved 2 degrees below 235 instead of above.
- B. The student used the closest number shown to 237.
- C. **Correct.** The student demonstrated an ability to use an analog thermometer to determine temperature.
- D. The student confused 227 and 237.

26 This clock shows the time Jordan's music class starts.



What time does Jordan's music class start?

- A** 2:40
- B** 3:08
- C** 3:40
- D** 8:13

Standard: 3.GM.3.1 Read and write time to the nearest five-minute interval (analog and digital).

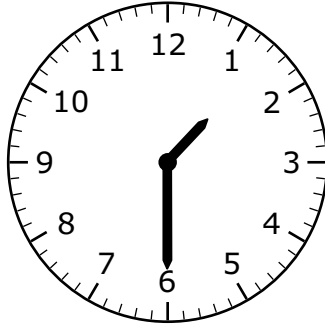
Depth-of-Knowledge: 1

This item is a DOK 1 because it requires the student to perform a simple procedure, identifying the time shown on an analog clock.

Distractor Rationale:

- A. Correct.** The student demonstrated an ability to read time to the nearest 5-minute.
- B.** The student chose because the minute hand is pointing to the 8 and the hour hand is closest to the 3.
- C.** The student chose because the hour hand is closer to the 3 than the 2.
- D.** The student swapped the hour and minute hands.

- 27** Ann's piano practice starts at 1:30 P.M. The practice ends 55 minutes later.



What time does the practice end?

- A** 12:35 P.M.
- B** 1:25 P.M.
- C** 1:55 P.M.
- D** 2:25 P.M.

Standard: 3.GM.3.2 Determine the solutions to problems involving addition and subtraction of time in intervals of 5 minutes, up to one hour, using pictorial models, number line diagrams, or other tools.

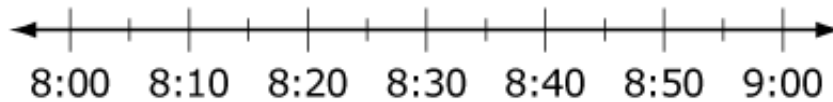
Depth-of-Knowledge: 2

This item is a DOK 2 because it requires the student to make a decision about how to find a time 55 minutes after a given time.

Distractor Rationale:

- A. The student found 55 minutes before 1:30 P.M.
- B. Balance distractor
- C. The student selected because 55 is given in the problem.
- D. Correct. The student demonstrated an ability to determine the solutions to a problem involving addition of time.

28 You may use the number line to help answer the question.



Students at a school begin art class at 8:15 A.M. The art class lasts for 35 minutes.

At what time does the art class end?

- A** 8:35 A.M.
- B** 8:40 A.M.
- C** 8:45 A.M.
- D** 8:50 A.M.

Standard: 3.GM.3.2 Determine the solutions to problems involving addition and subtraction of time in intervals of 5 minutes, up to one hour, using pictorial models, number line diagrams, or other tools.

Depth-of-Knowledge: 2

This item is a DOK 2 because it requires the student to make a decision about how to find a time 35 minutes after a given time. The student may use a model to help.

Distractor Rationale:

- A. The student knew the hour is 8 and thinks the minutes are 35 because 35 is given in the problem.
- B. The student added 25 minutes to 8:15.
- C. The student added 30 minutes to 8:15.
- D. **Correct.** The student demonstrated an ability to determine a solution to a problem involving the addition of time using a number line.

29 The table shows the ice-cream cones sold during lunch.

Ice-Cream Cones Sold

Flavor	Number of Cones
chocolate	5
strawberry	2
vanilla	4

Which pictograph shows the same information as the table?

A Ice-Cream Cones Sold

Flavor	Number of Cones
chocolate	▼▼▼▼▼
strawberry	▼▼
vanilla	▼▼▼▼

Key: ▼ = 2 cones

B Ice-Cream Cones Sold

Flavor	Number of Cones
chocolate	▼▼▼
strawberry	▼
vanilla	▼▼

Key: ▼ = 2 cones

C Ice-Cream Cones Sold

Flavor	Number of Cones
chocolate	▼▼
strawberry	▼
vanilla	▼

Key: ▼ = 2 cones

D Ice-Cream Cones Sold

Flavor	Number of Cones
chocolate	▼▼▼
strawberry	▼
vanilla	▼▼

Key: ▼ = 2 cones

29 continued...

Standard: 3.D.1.1 Collect and organize a data set with multiple categories using a frequency table, line plot, pictograph, or bar graph with scaled intervals.

Depth-of-Knowledge: 2

This item is a DOK 2 because it requires the student to translate information from a table into a pictograph with a key greater than 1.

Distractor Rationale:

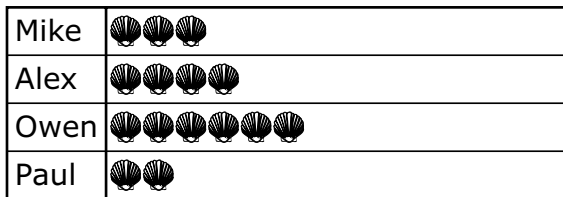
- A. The student ignored the key and used 1 cone picture to represent 1 cone.
- B. Correct. The student demonstrated an ability to represent data in a pictograph with scaled intervals.**
- C. Balance distractor
- D. The student did not know how to represent 5 cones with a key of 2.

30 Four friends collected seashells on the beach.

- Mike collected 6 seashells.
- Alex collected 2 more seashell than Mike.
- Owen collected 12 seashells.
- Paul and Owen together collected 16 seashells.

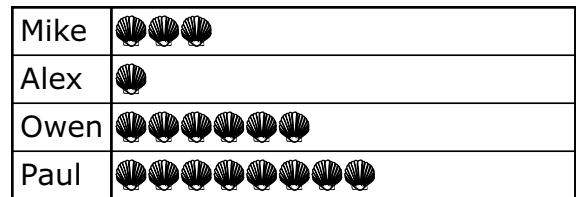
Which pictograph shows how many seashells each friend collected?

A Number of Seashells



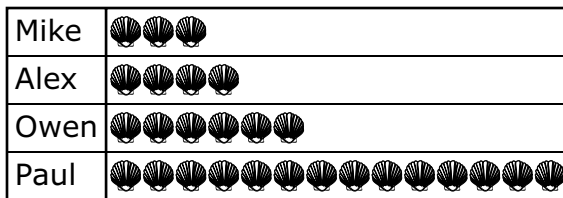
 = 2

B Number of Seashells



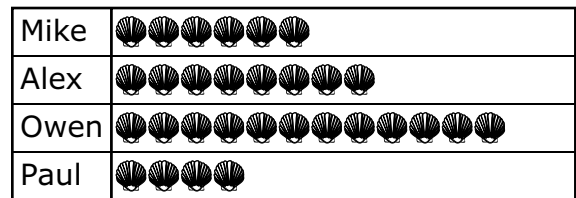
 = 2

C Number of Seashells



 = 2

D Number of Seashells



 = 2

Standard: 3.D.1.1 Collect and organize a data set with multiple categories using a frequency table, line plot, pictograph, or bar graph with scaled intervals.

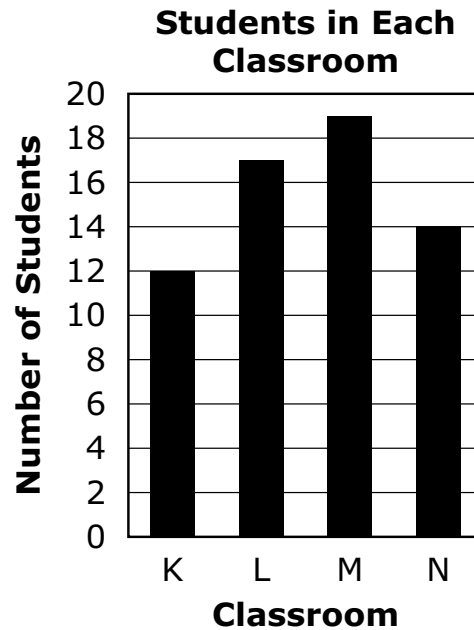
Depth-of-Knowledge: 2

This item is a DOK 2 because it requires the student to translate given information into a pictograph with a key greater than 1.

Distractor Rationale:

- A. **Correct.** The student demonstrated an ability to construct a pictograph with scaled intervals from a data set.
- B. The student used 2 for Alex and 16 for Paul because those are the numbers given in the bullets.
- C. The student saw correct for Mike, Alex, and Owen.
- D. The student did not use the key and treats each shell picture as 1 shell.

- 31** The graph shows the number of students in each of four classrooms.



How many more students are in classroom L than in classroom N?

- A** 3
- B** 4
- C** 5
- D** 7

Standard: 3.D.1.2 Solve one- and two-step problems using categorical data represented with a frequency table, pictograph, or bar graph with scaled intervals.

Depth-of-Knowledge: 2

This item is a DOK 2 because it requires the student to use a bar graph to answer a question about the data.

Distractor Rationale:

- A. Correct. The student demonstrated an understanding of how to solve a problem using categorical data presented in a bar graph.
- B. The student rounded classroom L up to 18 because that is the next closest labeled number.
- C. The student found the difference in the number of students in classroom M compared to classroom N.
- D. The student found the difference in the number of students in the two classrooms with the largest and smallest bars in the graph.

Cluster Items

The following sample items are part of a cluster. The cluster is presented first and then the two items that follow require use of the cluster. The two items are from different standards.

Use the information to answer the following questions.

Casey and her brother, Sam, keep track of their scores while playing their favorite video game. Sam's highest score is 5,400. Casey's scores for her last six games are shown.

Casey's Scores	
5,275	5,735
6,005	5,630
6,020	6,250

32 Sam's goal for next week is to score 1,000 more than his highest score. What is his goal?

- A 5,500
- B 6,400
- C 6,500
- D 7,250

Standard: 3.N.1.3 Applying knowledge of place values, use mental strategies (no written computations) to find 100 more or 100 less than a given number, 1,000 more or 1,000 less than a given number, and 10,000 more or 10,000 less than a given number, up to a five-digit number.

Depth-of-Knowledge: 1

This item is a DOK 1 because it requires the student to complete a simple procedure, finding 1,000 more than a given number.

Distractor Rationale:

- A. The student confused 1,000 and 100.
- B. Correct. The student demonstrated an ability to find 1,000 more than a given four-digit number.
- C. The student confused 1,000 and 1,100.
- D. The student confused Sam and Casey.

33 Casey wants to list her scores from greatest to least. Which list shows Casey's scores from greatest to least?

- A** 5,275; 5,630; 5,735; 6,005; 6,020; 6,250
- B** 5,735; 5,630; 5,275; 6,250; 6,020; 6,005
- C** 6,250; 6,005; 6,020; 5,275; 5,630; 5,735
- D** 6,250; 6,020; 6,005; 5,735; 5,630; 5,275

Standard: 3.N.1.4 Use place value to compare and order whole numbers up to 100,000, using comparative language, numbers, and symbols.

Depth-of-Knowledge: 1

This item is a DOK 1 because it requires the student to complete a simple task, ordering whole numbers from greatest to least.

Distractor Rationale:

- A. The student listed the scores from least to greatest instead of greatest to least.
- B. The student listed the scores in the five thousands from greatest to least and then the scores in the six thousands from greatest to least.
- C. The student mixed up 6,020 and 6,005 and 5,735 and 5,275.
- D. **Correct.** The student demonstrated an ability to use place value to compare whole numbers up to 100,000.